

# CARSON SILA WAMBUA

## OPERATIONAL DATA ANALYST | SRE ANALYTICS & OPERATIONS INTELLIGENCE

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### PROFESSIONAL SUMMARY

Results-driven **Operational Data Analyst** with deep domain experience bridging telecommunications infrastructure, Network Operations Center (NOC) environments, and machine learning operations intelligence. Proven track record at **Quavatel Limited** maintaining mission-critical SLA uptime for tier-1 telecom operators (Safaricom, Airtel, Liquid Telecom). Creator of **OpsPulse AI**, an enterprise operations intelligence suite that leverages unsupervised anomaly detection (Isolation Forest) and grounded runbook triage to reduce Mean Time to Resolution (MTTR) by **73%** and prevent up to **\$320,000** in annual contractual SLA penalties. Highly skilled in Python, SQL, Streamlit, predictive modeling, and executive KPI reporting.

### CORE TECHNICAL & OPERATIONAL COMPETENCIES

|                                |                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Operational Analytics:</b>  | SLA Compliance Monitoring, MTTR Optimization, Incident Blast Radius Scoring, Root Cause Analysis (5 Whys), SRE Telemetry, Fleet Tracking Analytics |
| <b>Languages &amp; Tools:</b>  | Python (Pandas, NumPy, Scikit-Learn, SciPy), SQL (PostgreSQL, MySQL), Bash/Shell, Git/GitHub, Great Expectations, Pytest                           |
| <b>Machine Learning:</b>       | Isolation Forest Anomaly Detection, Time-Series Diagnostics, Vibration Sensor Failure Prediction, Random Forest, Feature Engineering               |
| <b>Visualization &amp; BI:</b> | Streamlit Web Applications, Plotly Interactive Dashboards, Matplotlib, Seaborn, Automated Executive Briefing Generation                            |
| <b>Network &amp; Systems:</b>  | NOC Operations, Fiber-To-The-Home (FTTH), OTDR Fiber Cable Diagnostics, Transmission Restoration, Cloud Microservice Telemetry                     |

### PROFESSIONAL EXPERIENCE

**Operational Data Analyst Fellow — Enterprise SRE & Analytics Program** June 2026 – Present

**Power Learn Project (PLP) / Inuka Enterprise Analytics Track** Nairobi, Kenya

- Architected and deployed **OpsPulse AI**, an end-to-end operations copilot processing 1,800+ monthly incident logs to predict SLA breaches 15 minutes before expiration, cutting triage time from 45 min to 12 min (**-73% MTTR**).
- Modeled multi-tier financial impact for operational leadership, demonstrating a **461.4% Year 1 ROI** and **\$320,000** in annual savings across penalty avoidance, labor recovery, and churn mitigation.
- Implemented unsupervised machine learning models (**Isolation Forest**) over multivariate incident telemetry (duration, customer blast radius, error codes) to isolate operational anomalies with **94% precision**.
- Built automated 1-click executive daily briefing generators producing C-suite Markdown/CSV summaries, eliminating 2 hours of daily manual reporting.

**Network Operations & Telecommunications Engineering Intern** April 2024 – October 2024

**Quavatel Limited — Network Operations Center (NOC) & Field Engineering** Nairobi / Thika, Kenya

- Monitored 24/7 network telemetry and fiber transmission backbone for tier-1 telecom clients including **Safaricom, Airtel, and Liquid Telecom**, consistently sustaining a **99.8% SLA uptime** benchmark.
- Executed emergency cable cut localization and precision optical fiber splicing under strict contractual SLA recovery windows, accelerating average restoration time by **28%**.
- Maintained and audited enterprise GPS fleet tracking and asset telematics software, optimizing vehicle dispatch turnaround and verifying reporting uptime across 50+ field operational units.
- Conducted on-site Fiber-To-The-Home (FTTH) network installations, optical power testing, and customer premise troubleshooting, raising first-visit resolution to **92%**.

### STRATEGIC DATA ANALYTICS PROJECTS

**OpsPulse AI: Enterprise Operations Intelligence & SLA Copilot** Python, Streamlit, Scikit-Learn, Plotly

- Built an interactive Streamlit operations hub featuring real-time KPI monitors, interactive Plotly breach heatmaps, and dynamic severity segmentation for Site Reliability Engineering (SRE) workflows.
- Engineered a dual-mode AI triage engine combining grounded RAG knowledge search with an offline heuristic fallback, autogenerating 5-Whys root cause post-mortems and CLI remediation steps.
- Developed 13 automated unit tests with Pytest covering edge cases, missing data handling, and statistical bounds, ensuring 100% test suite reliability.

**Predictive Equipment Vibration & Machine Failure Detection System** Python, Scikit-Learn, Pandas, Seaborn

- Engineered an operational machine learning model predicting mechanical motor failure from multi-axis vibration sensor streams 48 hours in advance, achieving a **0.94 ROC-AUC** score.
- Built Fourier transform (FFT) frequency feature extractors, cutting unplanned factory downtime by an estimated **35%**.

**Automated Telecom Telemetry & SRE Data Pipeline**

Python, PostgreSQL, Great Expectations

- Designed an automated ETL ingestion pipeline processing 50,000+ daily network telemetry records, with automated schema validation and SLA breach threshold alerting.

**EDUCATION**

**Bachelor of Science in Information Technology**

2024 – Nov 2026 (Expected)

Zetech University — Coursework Completed, Focus on Data Analytics & Networks

Nairobi, Kenya

**Diploma in Information Technology**

2022 – 2024

Zetech University — Graduated with Distinction; Created Centralized Examination Portal

Nairobi, Kenya

**HONORS, LEADERSHIP & VOLUNTEERING**

- **1st Runner-Up (2nd Place Overall)**, Annual University Innovation Hackathon (2025) — Led architecture for campus fintech solution.
- **Youth Ministry & Ushering Operations Leader**, CITAM Thika (2021 – 2025) — Managed weekly operational logistics, team scheduling, and hospitality for 500+ attendees.
- **E-Commerce Platform for Agricultural Trade** — Designed degree capstone portal connecting rural farmers with direct markets, earning highest honors from university examination panel.